

Interaction Topology in TDA

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Fall 2025 MTH994:

Mathematics-Assisted AI and AI-inspired Mathematics

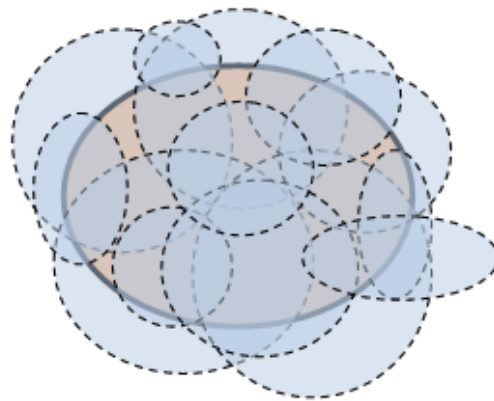
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History

- Interaction cohomology of forward and backward self-similar systems
 - Hiroki Sumi; Research Institute for Mathematical Sciences; 2009
 - Forward systems: (L, h_i) where $h_i: L \rightarrow L$ continuous map such that $L = \bigcup h_i(L)$
 - Introduce new invariant
- The cohomology for Wu characteristic
 - Oliver Knill; arXiv; 2018
 - Different construction for interaction cohomology on finite simplicial complexes
 - Connection to Wu characteristic
- Interaction homotopy and interaction homology
 - Jian Liu, Dong Chen, Guo-Wei Wei; arXiv; 2023
 - Introduce concept of interaction spaces and build homotopy and homology theories
- Persistent Interaction Topology in Data Analysis
 - Jian Liu, Dong Chen, Guo-Wei Wei; Foundation of Data Science; 2024
 - Build TDA tool for interaction spaces

Interaction space

- An interaction space $(X, \{X_i\}_{1 \leq i \leq k})$ is a topological space with covering.
- A morphism of n-interaction spaces is a family of continuous maps of topological spaces $f = \{f_i\}_{1 \leq i \leq k} X \rightarrow Y$ so that $f_i: X_i \rightarrow Y_i$ satisfying:
 - $f_i(x_i) = f_j(x_j) \iff x_i = x_j$ where $x_i \in X_i, x_j \in X_j$
- The condition enforces structure-preserving maps
 - $f_i(X_i \setminus X_j) \subset Y_i \setminus Y_j$
 - $f_i(X_i \cap X_j) \subset Y_i \cap Y_j$ and $f_j(X_i \cap X_j) \subset Y_i \cap Y_j$



Open cover

Interaction Homotopy

Definition 2.1. Let $f, g : (X, \{X_i\}_{1 \leq i \leq n}) \rightarrow (Y, \{Y_i\}_{1 \leq i \leq n})$ be morphisms of interaction spaces.

We say that f and g are formal homotopic, denoted by $f \simeq g$, if there exists a morphism of interaction spaces $H : (X \times I, \{X_i \times I\}_{1 \leq i \leq n}) \rightarrow (Y, \{Y_i\}_{1 \leq i \leq n})$ such that $H(x, 0) = f$ and $H(x, 1) = g$ for all $x \in X$.

We say f is homotopic to g if there are a family of formal homotopies $f \simeq f_1 \simeq \cdots \simeq f_k \simeq g$ connecting f and g , denoted by $f \sim g$.

Definition 2.2. A morphism of interaction spaces $f : (X, \{X_i\}_{1 \leq i \leq n}) \rightarrow (Y, \{Y_i\}_{1 \leq i \leq n})$ is a homotopy equivalence if there exists a morphism of interaction spaces $g : (Y, \{Y_i\}_{1 \leq i \leq n}) \rightarrow (X, \{X_i\}_{1 \leq i \leq n})$ such that $f \circ g \sim \text{id}|_{(Y, \{Y_i\}_{1 \leq i \leq n})}$ and $g \circ f \sim \text{id}|_{(X, \{X_i\}_{1 \leq i \leq n})}$.

Observation 1: Not transitive

Formal homotopy is

- Reflective
- Symmetric
- Not transitive

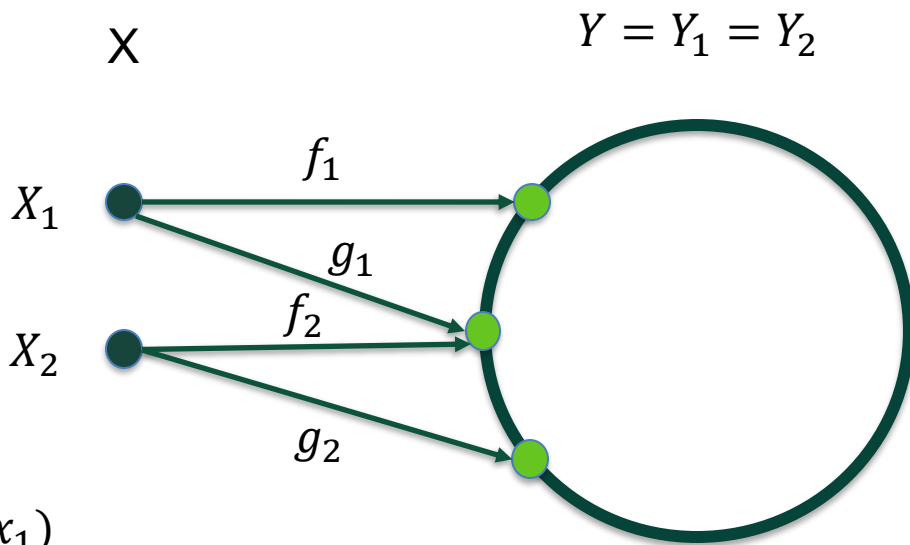
If $f = \{f_1, f_2\} \simeq g = \{g_1, g_2\}$, then

$$H_1(x_1, 0) = f_1(x_1), H_1(x_1, 1) = g_1(x_1)$$

$$H_2(x_2, 0) = f_2(x_2), H_2(x_2, 1) = g_2(x_2)$$

$$H_1(x_1, 1) = g_1(x_1) = f_2(x_2) = H_2(x_2, 0) \Rightarrow x_1 = x_2$$

$$f \not\simeq g$$

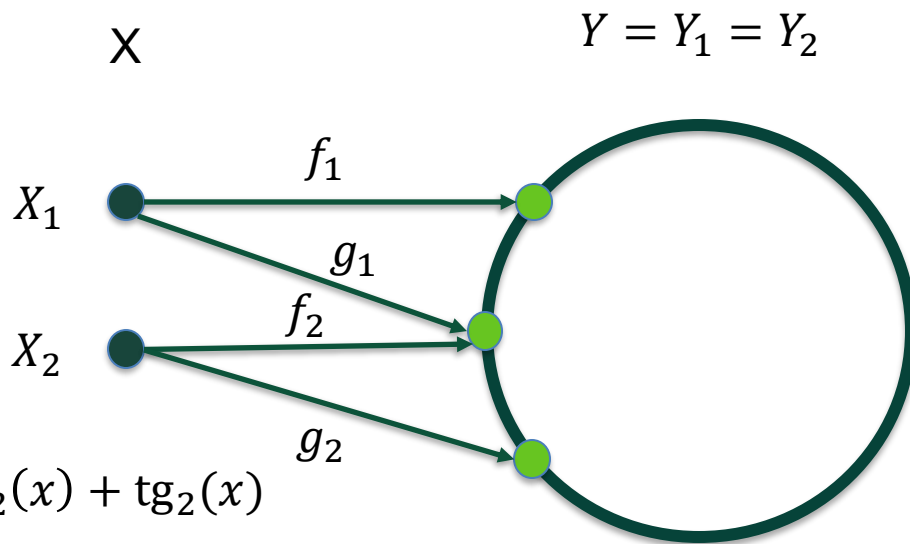


Observation 1: Not transitive

Formal homotopy is

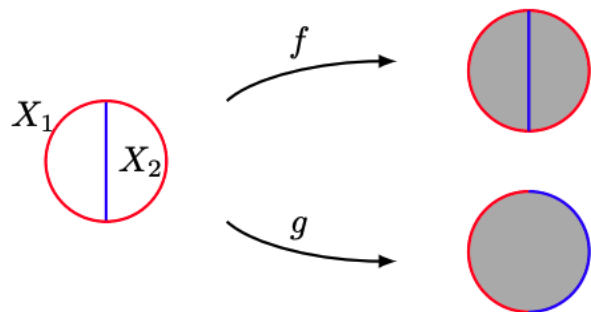
- Reflective
- Symmetric
- Not transitive

Let $f = \{f_1, f_2\}, h = \{f_1, g_2\}$, then
 $F_1(x, t) = f_1(x), F_2(x, t) = ((1 - t)f_2(x) + tg_2(x))$
 $\Rightarrow f \simeq h$
 Let $g = \{g_1, g_2\}, h = \{f_1, g_2\}$, then
 $G_1(x, t) = ((1 - t)g_1(x) + tf_1(x)), F_2(x, t) = g_2(x)$
 $\Rightarrow h \simeq g$
 $\Rightarrow f \simeq h \simeq g \Rightarrow f \sim g$ but $f \neq g$



Observation 2:

Let $f, g : (X, \{X_i\}_{1 \leq i \leq n}) \rightarrow (Y, \{Y_i\}_{1 \leq i \leq n})$ be morphisms of interaction spaces. If $f \simeq g : (X, \{X_i\}_{1 \leq i \leq n}) \rightarrow (Y, \{Y_i\}_{1 \leq i \leq n})$, then one has $f_i \simeq g_i : X_i \rightarrow Y_i$ for each $1 \leq i \leq n$. However, the converse is not always true. Indeed, the homotopy $H_i : X_i \times I \rightarrow Y_i$ from $f_i : X_i \rightarrow Y_i$ to $g_i : X_i \rightarrow Y_i$ for $1 \leq i \leq n$ does not always extend to a morphism of interaction spaces.



$$\begin{aligned} f_1(x, y) &= (x, y), f_2(0, y) = (0, y), \\ g_1(x, y) &= (-|x|, y), g_2(0, y) = (\sqrt{1 - y^2}, y). \end{aligned} \Rightarrow f_1 \simeq g_1, f_2 \simeq g_2$$

$$f \simeq g \Rightarrow H = \{H_1, H_2\} \text{ s.t.}$$

$$H_1(u, 0) = f_1(u), H_1(u, 1) = g_1(u)$$

$$H_2(v, 0) = f_2(v), H_2(v, 1) = g_2(v)$$

$$\text{Let } u = (1, 0), v = (0, 0)$$

$$H_1(u, 0) = (1, 0), H_1(u, 1) = (-1, 0)$$

$$H_2(v, 0) = (0, 0), H_2(v, 1) = (1, 0)$$

$$H_1(u, 0) = H_2(v, 1)$$

Standard Singular Homology vs Interaction Singular Homology

- X topological space
- p -simplex : $\sigma_p: \Delta^p \rightarrow X$
- Singular Chain complex:
 - $S_*(X): \cdots S_2(X) \xrightarrow{d_2} S_1(X) \xrightarrow{d_1} S_0(X) \xrightarrow{d_0} 0$
 - $d_n: S_n(X) \rightarrow S_{n-1}(X)$
 - $d_n = \sum_{i=0}^n (-1)^i \partial_i^n$
 - $\partial_i^n([v_0, \cdots, v_n]) = [v_0, \cdots, \hat{v}_i, \cdots, v_n]$
- Singular homology:
 - $H_n(X) = Z_n(X) / B_n(X)$
- $(X, \{X_i\}_{1 \leq i \leq k})$ an interaction space
- $S_*(X_i) \Rightarrow \bigotimes_i S_*(X_i)$ chain complex
- The interaction $(p_1, \cdots, p_k)$ -simplex $(\sigma_1, \cdots, \sigma_k)$
 - $\sigma_i: \Delta^{p_i} \rightarrow X_i$
 - $\bigcap_{i=1}^k \text{im} \sigma_i \neq \emptyset$
 - $d(\sigma_1 \otimes \cdots \otimes \sigma_k) = \sum_{i=1}^k (-1)^{\dim \sigma_1 + \cdots + \dim \sigma_{i-1}} \sigma_1 \otimes \cdots \otimes d\sigma_i \otimes \cdots \otimes \sigma_k$
- Singular Interaction Homology:
 - $H_n(\{X_i\}_i) = H_n(IS_*(\{X_i\}_i))$
 - Where $IS_*(\{X_i\}_i)$ is interaction singular chain complex

Interaction Singular Complex

$$\begin{array}{c}
 S_k(X_1) \quad S_k(X_2) \\
 \downarrow \quad \downarrow \\
 \vdots \quad \vdots \\
 S_2(X_1) \quad S_2(X_2) \\
 \downarrow d_2 \quad \downarrow d'_2 \\
 S_1(X_1) \quad S_1(X_2) \\
 \downarrow d_1 \quad \downarrow d'_1 \\
 S_0(X_1) \quad S_0(X_2) \\
 \downarrow d_0 \quad \downarrow d'_0 \\
 0 \quad 0
 \end{array}$$

$(X, \{X_i\}_{1 \leq i \leq k})$ an
interaction space

$$\begin{array}{c}
 \oplus_{i+j=k} S_i(X_1) \otimes S_j(X_2) \\
 \downarrow \quad \vdots \\
 S_0(X_1) \otimes S_2(X_2) \otimes (S_1(X_1) \otimes S_1(X_2)) \otimes (S_2(X_1) \otimes S_0(X_2)) \\
 \downarrow \\
 (S_0(X_1) \otimes S_1(X_2)) \otimes (S_1(X_1) \otimes S_0(X_2)) \\
 \downarrow \\
 S_0(X_1) \otimes S_0(X_2) \\
 \downarrow \\
 0
 \end{array}$$

Tensor product of all complexes
 $\otimes S_*(X_i)$

$$\begin{array}{c}
 \oplus_{i+j=k} S_i(X_1) \otimes S_j(X_2) / T_k(X_1, X_2) \\
 \downarrow \quad \vdots \\
 (\oplus_{i+j=2} S_i(X_1) \otimes S_j(X_2)) / T_2(X_1, X_2) \\
 \downarrow \\
 (\oplus_{i+j=2} S_i(X_1) \otimes S_j(X_2)) / T_1(X_1, X_2) \\
 \downarrow \\
 (\oplus_{i+j=2} S_i(X_1) \otimes S_j(X_2)) / T_0(X_1, X_2) \\
 \downarrow \\
 0
 \end{array}$$

Submodule $T_*(\{X_i\}_i)$ where $\cap \sigma_i = \emptyset$
 $d(T_*(\{X_i\}_i)) \subset T_*(\{X_i\}_i)$
 Interaction Chain Complex
 $IS_*(\{X_i\}_i) = \otimes S_*(X_i) / T_*(\{X_i\}_i)$

Interaction homology and homology of a total complex

Definition. A **bicomplex** (or **double complex**) is an ordered triple (M, d', d'') , where $M = (M_{p,q})$ is a bigraded module, $d', d'': M \rightarrow M$ are differentials of bidegree $(-1, 0)$ and $(0, -1)$, respectively (so that $d'd' = 0$ and $d''d'' = 0$), and

$$d'_{p,q-1}d''_{p,q} + d''_{p-1,q}d'_{p,q} = 0.$$

Definition. If M is a bicomplex, then its **total complex**, denoted by $\text{Tot}(M)$, is the complex with n th term

$$\text{Tot}(M)_n = \bigoplus_{p+q=n} M_{p,q}$$

and with differentials $D_n: \text{Tot}(M)_n \rightarrow \text{Tot}(M)_{n-1}$ given by

$$D_n = \sum_{p+q=n} (d'_{p,q} + d''_{p,q})$$

In our case $M_{p,q} = S_p(X_1) \otimes S_q(X_2)$

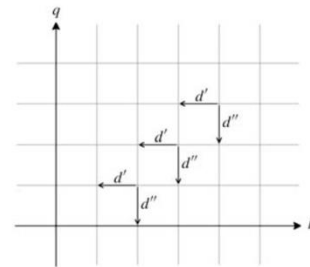


Fig. 10.1 Bicomplex.

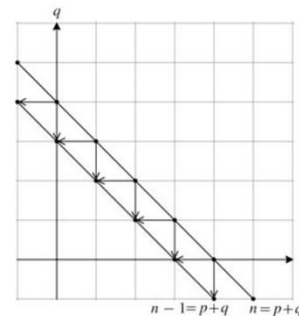
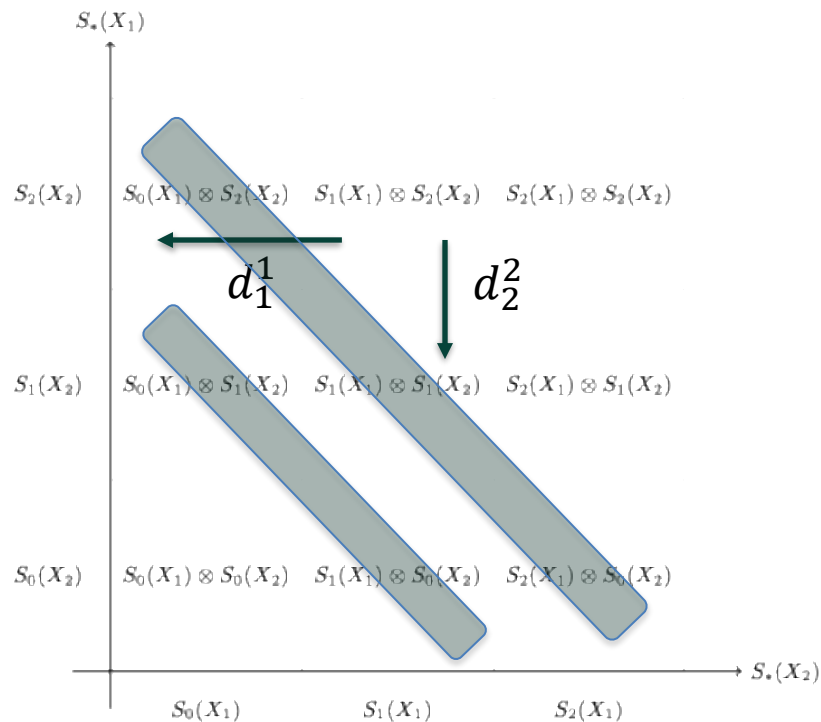


Fig. 10.2 Total complex.

Interaction homology and homology of a total complex



- Total complex for first quadrant bi-complex is the tensor product

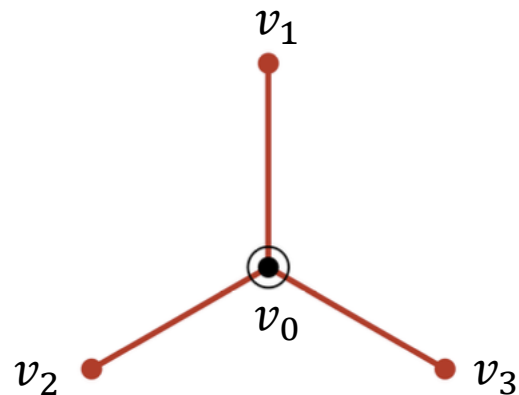
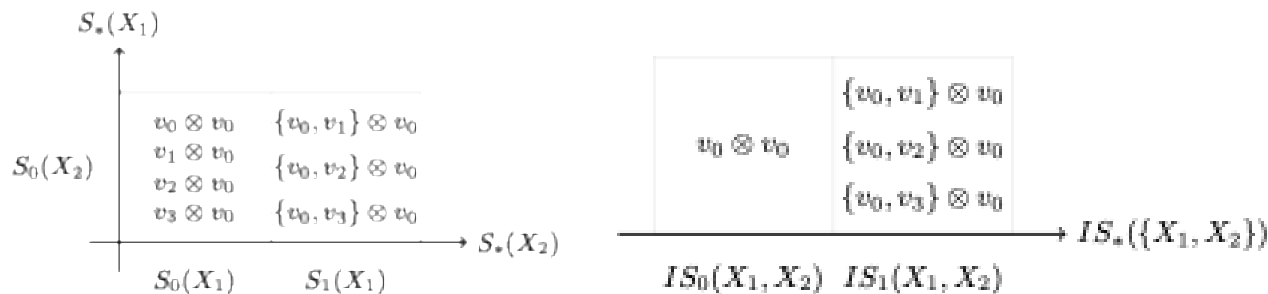
$$D_n: a_p \otimes b_q \mapsto \Delta' a_p \otimes b_q + (-1)^p a_p \otimes \Delta'' b_q$$

- Collapse a bi-graded into a single complex
- In generalized version, the following helps us to reduce the information into single complex

$$d(\sigma_1 \otimes \cdots \otimes \sigma_k) = \sum_{i=1}^k (-1)^{\dim \sigma_1 + \cdots + \dim \sigma_{i-1}} \sigma_1 \otimes \cdots \otimes d\sigma_i \otimes \cdots \otimes \sigma_k$$

Example on simplicial complex

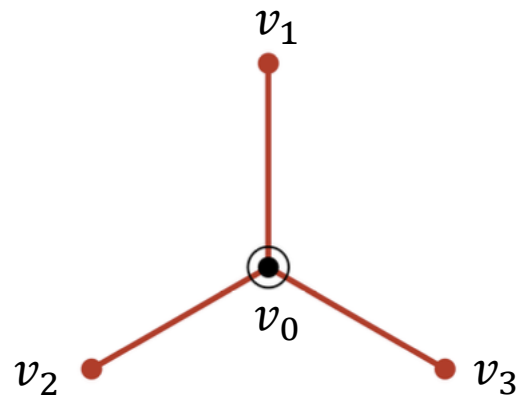
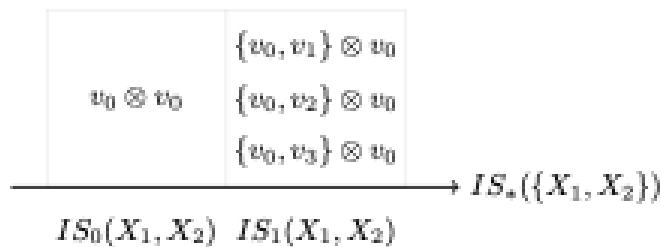
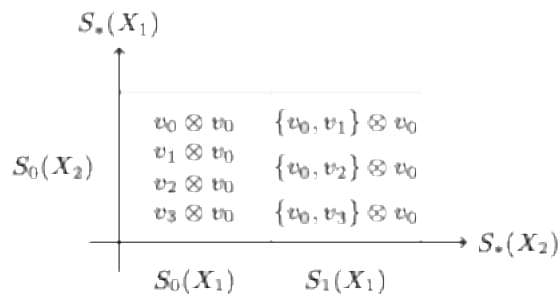
$$X_1 = \{\{v_0\}, \{v_1\}, \{v_2\}, \{v_3\}, \{v_0, v_1\}, \{v_0, v_2\}, \{v_0, v_3\}\}, \quad X_2 = \{\{v_0\}\}.$$



$$\begin{aligned}
 d_1(\{v_0, v_1\} \otimes v_0) &= d_1^1(\{v_0, v_1\}) \otimes v_0 + (-1)^1 \{v_0, v_1\} \otimes d_0^2(v_0) \\
 &= (v_1 - v_0) \otimes v_0 + (-1)^1 \{v_0, v_1\} \otimes 0 \\
 &= v_1 \otimes v_0 - v_0 \otimes v_0 = -v_0 \otimes v_0
 \end{aligned}$$

Example on simplicial complex

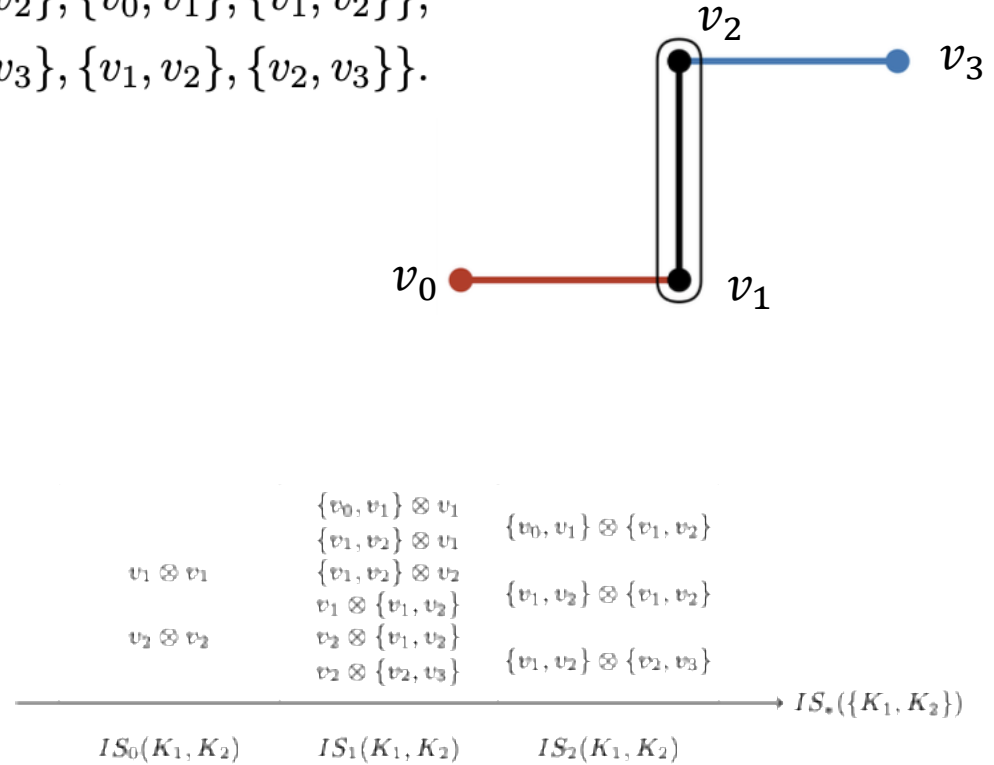
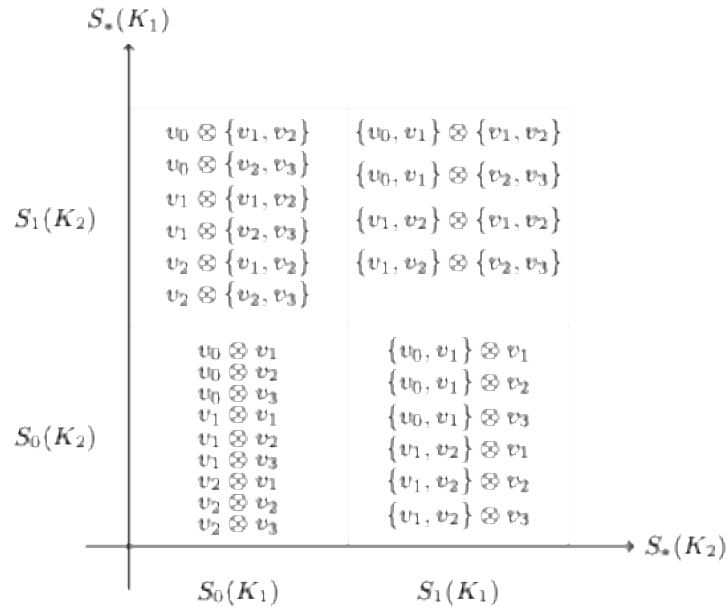
$$X_1 = \{\{v_0\}, \{v_1\}, \{v_2\}, \{v_3\}, \{v_0, v_1\}, \{v_0, v_2\}, \{v_0, v_3\}\}, \quad X_2 = \{\{v_0\}\}.$$



$$d_1 = v_0 \otimes v_0 \begin{pmatrix} -1 & -1 & -1 \end{pmatrix}$$

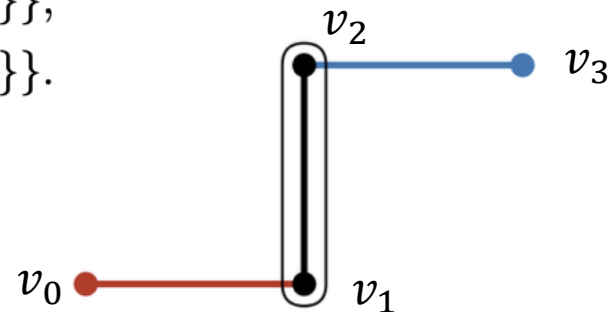
$$\begin{aligned} H_1(IS_*) &= \ker d_1 / \text{im } d_0 \\ &= \langle \{v_0, v_2\} \otimes v_0 - \{v_0, v_1\} \otimes v_0, \\ &\quad \{v_0, v_3\} \otimes v_0 - \{v_0, v_1\} \otimes v_0 \rangle \\ &= \mathbb{K} \oplus \mathbb{K} \end{aligned}$$

Example 2 $K_1 = \{\{v_0\}, \{v_1\}, \{v_2\}, \{v_0, v_1\}, \{v_1, v_2\}\},$
 $K_2 = \{\{v_1\}, \{v_2\}, \{v_3\}, \{v_1, v_2\}, \{v_2, v_3\}\}.$



Example 2 $K_1 = \{\{v_0\}, \{v_1\}, \{v_2\}, \{v_0, v_1\}, \{v_1, v_2\}\},$
 $K_2 = \{\{v_1\}, \{v_2\}, \{v_3\}, \{v_1, v_2\}, \{v_2, v_3\}\}.$

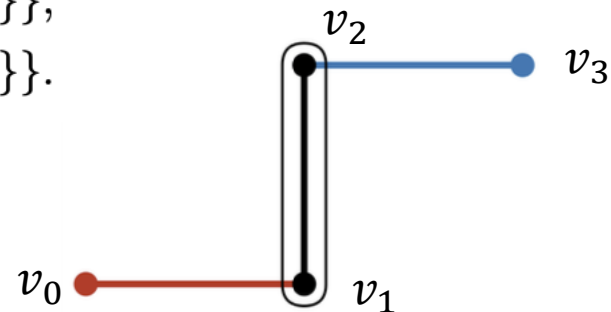
	$\{v_0, v_1\} \otimes v_1$	$\{v_0, v_1\} \otimes \{v_1, v_2\}$	
$v_1 \otimes v_1$	$\{v_1, v_2\} \otimes v_1$		
	$\{v_1, v_2\} \otimes v_2$	$\{v_1, v_2\} \otimes \{v_1, v_2\}$	
$v_2 \otimes v_2$	$v_1 \otimes \{v_1, v_2\}$		
	$v_2 \otimes \{v_1, v_2\}$	$\{v_1, v_2\} \otimes \{v_2, v_3\}$	
	$v_2 \otimes \{v_2, v_3\}$		
$\xrightarrow{IS_*(\{K_1, K_2\})}$			
$IS_0(K_1, K_2)$	$IS_1(K_1, K_2)$	$IS_2(K_1, K_2)$	



$$\begin{aligned}
 d_1(\{v_0, v_1\} \otimes v_1) &= d_1^1(\{v_0, v_1\}) \otimes v_1 + (-1)^1 \{v_0, v_1\} \otimes d_0^2(v_1) \\
 &= (v_1 - v_0) \otimes v_1 + (-1)^1 \{v_0, v_1\} \otimes 0 \\
 &= v_1 \otimes v_1 - v_0 \otimes v_1 = v_1 \otimes v_1
 \end{aligned}$$

Example 2 $K_1 = \{\{v_0\}, \{v_1\}, \{v_2\}, \{v_0, v_1\}, \{v_1, v_2\}\},$
 $K_2 = \{\{v_1\}, \{v_2\}, \{v_3\}, \{v_1, v_2\}, \{v_2, v_3\}\}.$

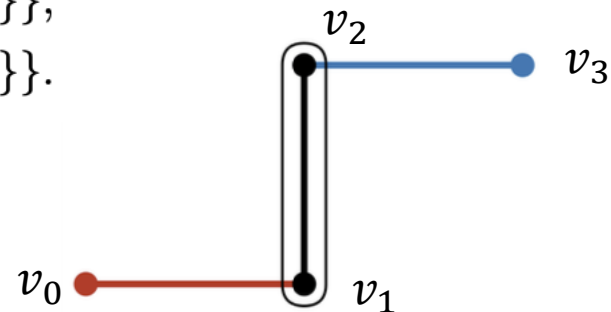
	$\{v_0, v_1\} \otimes v_1$	$\{v_0, v_1\} \otimes \{v_1, v_2\}$	
$v_1 \otimes v_1$	$\{v_1, v_2\} \otimes v_1$		
	$\{v_1, v_2\} \otimes v_2$	$\{v_1, v_2\} \otimes \{v_1, v_2\}$	
$v_2 \otimes v_2$	$v_1 \otimes \{v_1, v_2\}$		
	$v_2 \otimes \{v_1, v_2\}$	$\{v_1, v_2\} \otimes \{v_2, v_3\}$	
	$v_2 \otimes \{v_2, v_3\}$		
			$\rightarrow IS_*(\{K_1, K_2\})$
$IS_0(K_1, K_2)$	$IS_1(K_1, K_2)$	$IS_2(K_1, K_2)$	



$$\begin{aligned}
 d_2(\{v_0, v_1\} \otimes \{v_1, v_2\}) &= d_1^1(\{v_0, v_1\}) \otimes \{v_1, v_2\} + (-1)^1 \{v_0, v_1\} \otimes d_1^2(\{v_1, v_2\}) \\
 &= (v_1 - v_0) \otimes \{v_1, v_2\} + (-1)^1 \{v_0, v_1\} \otimes (v_2 - v_1) \\
 &= v_1 \otimes \{v_1, v_2\} - v_0 \otimes \{v_1, v_2\} - \{v_0, v_1\} \otimes v_2 + \{v_0, v_1\} \otimes v_1 \\
 &= \{v_0, v_1\} \otimes v_1 + v_1 \otimes \{v_1, v_2\}
 \end{aligned}$$

Example 2 $K_1 = \{\{v_0\}, \{v_1\}, \{v_2\}, \{v_0, v_1\}, \{v_1, v_2\}\},$
 $K_2 = \{\{v_1\}, \{v_2\}, \{v_3\}, \{v_1, v_2\}, \{v_2, v_3\}\}.$

	$\{v_0, v_1\} \otimes v_1$	$\{v_0, v_1\} \otimes \{v_1, v_2\}$	
	$\{v_1, v_2\} \otimes v_1$		
$v_1 \otimes v_1$	$\{v_1, v_2\} \otimes v_2$	$\{v_1, v_2\} \otimes \{v_1, v_2\}$	
	$v_1 \otimes \{v_1, v_2\}$		
$v_2 \otimes v_2$	$v_2 \otimes \{v_1, v_2\}$	$\{v_1, v_2\} \otimes \{v_2, v_3\}$	
	$v_2 \otimes \{v_2, v_3\}$		
$\xrightarrow{IS_*(\{K_1, K_2\})}$			
$IS_0(K_1, K_2)$	$IS_1(K_1, K_2)$	$IS_2(K_1, K_2)$	



$$d_1 = \begin{matrix} v_1 \otimes v_1 \\ v_2 \otimes v_2 \end{matrix} \begin{bmatrix} 1 & -1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 \end{bmatrix}$$

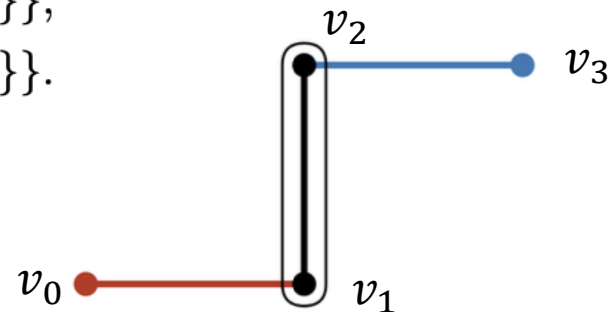
$$d_2 = \begin{matrix} \{v_0, v_1\} \otimes v_1 \\ \{v_1, v_2\} \otimes v_1 \\ \{v_1, v_2\} \otimes v_2 \\ v_1 \otimes \{v_1, v_2\} \\ v_2 \otimes \{v_1, v_2\} \\ v_2 \otimes \{v_2, v_3\} \end{matrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example 2 $K_1 = \{\{v_0\}, \{v_1\}, \{v_2\}, \{v_0, v_1\}, \{v_1, v_2\}\},$
 $K_2 = \{\{v_1\}, \{v_2\}, \{v_3\}, \{v_1, v_2\}, \{v_2, v_3\}\}.$

	$\{v_0, v_1\} \otimes v_1$	$\{v_0, v_1\} \otimes \{v_1, v_2\}$	
	$\{v_1, v_2\} \otimes v_1$		
$v_1 \otimes v_1$	$\{v_1, v_2\} \otimes v_2$	$\{v_1, v_2\} \otimes \{v_1, v_2\}$	
	$v_1 \otimes \{v_1, v_2\}$		
$v_2 \otimes v_2$	$v_2 \otimes \{v_1, v_2\}$	$\{v_1, v_2\} \otimes \{v_2, v_3\}$	
	$v_2 \otimes \{v_2, v_3\}$		

$\xrightarrow{\quad\quad\quad} IS_*(\{K_1, K_2\})$

$IS_0(K_1, K_2)$
 $IS_1(K_1, K_2)$
 $IS_2(K_1, K_2)$



$$d_1 = \begin{matrix} v_1 \otimes v_1 \\ v_2 \otimes v_2 \end{matrix} \begin{bmatrix} 1 & -1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 \end{bmatrix}$$

$$H_0 = \ker d_0 / \text{imd}_1 = 0 \text{ rank is 2.}$$

$$H_1 = \ker d_1 / \text{imd}_2 = \langle \{v_0, v_1\} \otimes v_1 + \{v_1, v_2\} \otimes v_1 \rangle$$

$$H_2 = \ker d_2 / \text{imd}_3 = 0 \text{ rank is 3.}$$

$$d_2 = \begin{matrix} \{v_0, v_1\} \otimes v_1 \\ \{v_1, v_2\} \otimes v_1 \\ \{v_1, v_2\} \otimes v_2 \\ v_1 \otimes \{v_1, v_2\} \\ v_2 \otimes \{v_1, v_2\} \\ v_2 \otimes \{v_2, v_3\} \end{matrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Interaction Betti Numbers

Corollary 3.4. *Let $f : \{X_i\}_{1 \leq i \leq n} \rightarrow \{Y_i\}_{1 \leq i \leq n}$ be a homotopy equivalence between interaction spaces. Then $H_p(f)$ is an isomorphism for any $p \geq 0$.*

Betti numbers are the dimension of the homology groups on the corresponding dimension

$$\beta_n^k(\{X_i\}_{1 \leq i \leq k}) = \dim(H_n(IS_*(\{X_i\}_{1 \leq i \leq k})))$$

$$\beta_0^2(\{X_1, X_2\}) = 0$$

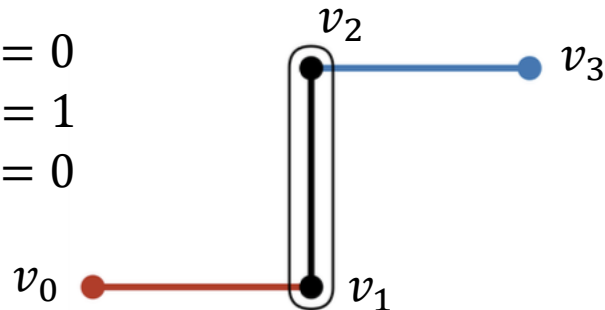
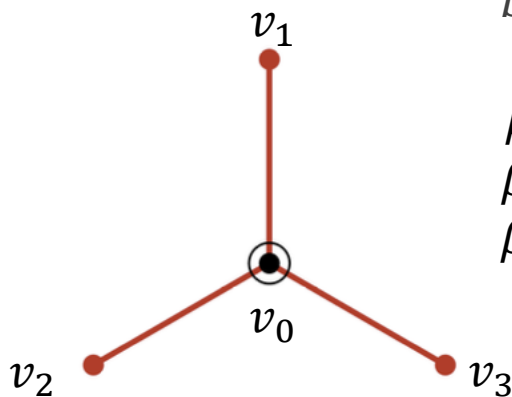
$$\beta_1^2(\{X_1, X_2\}) = 2$$

$$\beta_2^2(\{X_1, X_2\}) = 0$$

$$\beta_0^2(\{K_1, K_2\}) = 0$$

$$\beta_1^2(\{K_1, K_2\}) = 1$$

$$\beta_2^2(\{K_1, K_2\}) = 0$$



Euler Char and Wu characteristic

1. Euler char :

- $\chi = \sum_n (-1)^n \dim(C_n) = \sum_n (-1)^n \dim(H_n)$

2. Wu Char :

- $\omega(G) = \sum_{x \sim y} \omega(x) \omega(y)$ where $x \sim y$ means simplex x intersect with simplex y

- $\omega(x) = (-1)^{\dim x}$

- $\omega(G \times H) = \omega(G) \omega(H)$

- $\omega(G \cup H) = \omega(G) + \omega(H) - \omega(G \cap H)$

3. Higher order Wu char

- $\omega_k(G) = \sum_{x_1 \sim x_2 \sim \dots \sim x_k} \prod_{t=1}^k \omega(x_t)$

- $\omega_1(G) = \chi(G)$

- $\omega_2(G) = \sum_{p \geq 0} (-1)^p \beta_p^2(\{G, G\})$ where β_p^2 is the p th interaction betti number

Wu char of 2-simplex

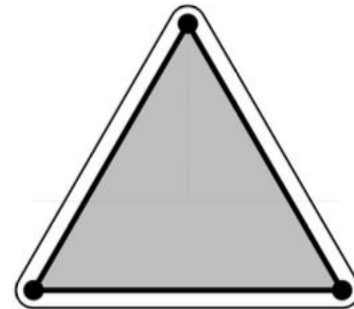
$$\partial\Delta[2] \quad \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}.$$

$$(\{0\}, \{0, 1\}), (\{0\}, \{0, 2\}), (\{1\}, \{0, 1\}), (\{1\}, \{1, 2\}), (\{2\}, \{0, 2\}), (\{2\}, \{1, 2\}),$$

$$(\{0, 1\}, \{0\}), (\{0, 1\}, \{1\}), (\{0, 2\}, \{0\}), (\{0, 2\}, \{2\}), (\{1, 2\}, \{1\}), (\{1, 2\}, \{2\}).$$

$$(\{0, 1\}, \{0, 1\}), (\{0, 1\}, \{0, 2\}), (\{0, 1\}, \{1, 2\}), (\{0, 2\}, \{0, 1\}), (\{0, 2\}, \{0, 2\}),$$

$$(\{0, 2\}, \{1, 2\}), (\{1, 2\}, \{0, 1\}), (\{1, 2\}, \{0, 2\}), (\{1, 2\}, \{1, 2\}).$$



$$\omega(\partial\Delta[2]) = \sum_{\sigma \sim \tau} (-1)^{\dim \sigma + \dim \tau} = 3 - 12 + 9 = 0.$$

$$\omega(\partial\Delta[2]) = \sum_{i \geq 0} (-1)^i \beta_{2,i}(\partial\Delta[2]) = 0 - 1 + 1 = 0.$$

$$\omega(\Delta[2]) = \sum_{\sigma \sim \tau} (-1)^{\dim \sigma + \dim \tau} = 3 - 12 + 15 - 6 + 1 = 1.$$

$$\omega(\Delta[2]) = \sum_{i \geq 0} (-1)^i \beta_{2,i}(\Delta[2]) = 1.$$

How to compute Interaction Betti Numbers?

Let B_p be the matrix representation of the differential

$$d_p: IS_d(\{X_i\}_{1 \leq i \leq k}) \rightarrow IS_{d-1}(\{X_i\}_{1 \leq i \leq k})$$

$$\beta_p^k(\{X_i\}_{1 \leq i \leq k}) = \dim(IS_p) - \text{rank } B_p - \text{rank } B_{p+1}$$

Persistent Interaction Homology

Interaction homology H_* is a functor from InteractionChains to Vec

Let $IC_*(K_a)$ be an interaction chain complex for each a so that

$$\text{Filtration:} \quad IC_*(K_{a_1}) \subset IC_*(K_{a_2}) \subset \cdots \subset IC_*(K_{a_m})$$

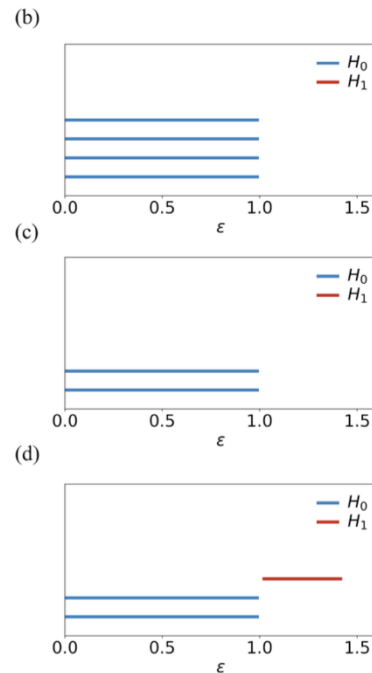
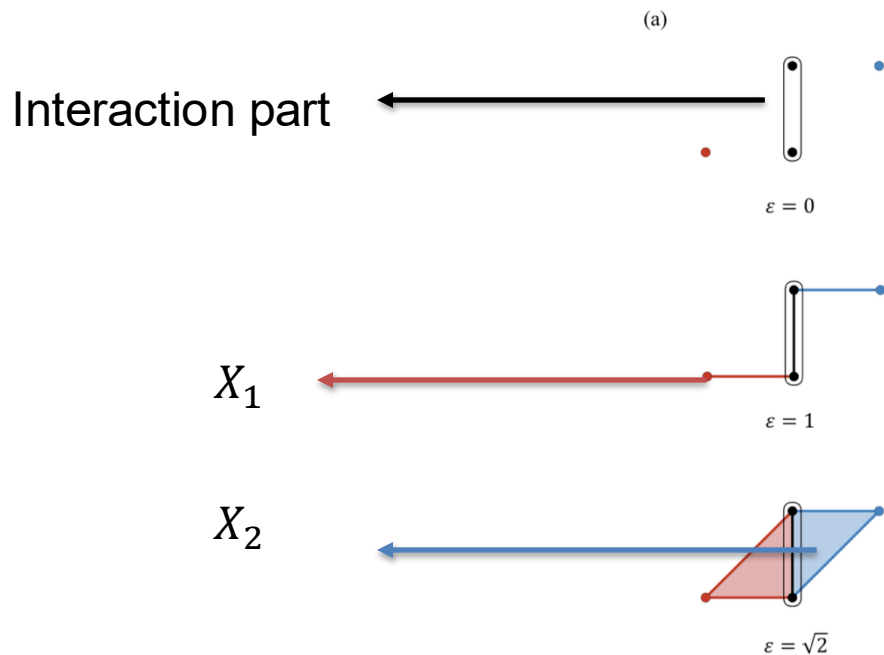
$$\text{Induced maps: } H_*(IC_*(K_{a_1})) \subset H_*(IC_*(K_{a_2})) \subset \cdots \subset H_*(IC_*(K_{a_m}))$$

- Persistent classes (a,b)-persistent Mayer homology

$$H_{n,(a,b)} := \text{im}\{H_n(IC_*(K_a)) \rightarrow H_n(IC_*(K_b))\}$$

- The rank of $H_{n,(a,b)}$ is the interaction (a,b)-persistent Betti numbers.

Persistent Interaction Barcodes



$$X_1 \cup X_2$$

$$X_1 \cap X_2$$

$$IC_*({X_1, X_2})$$

Interaction Laplacian

$$\langle \sigma_1 \otimes \cdots \otimes \sigma_n, \tau_1 \otimes \cdots \otimes \tau_n \rangle = \prod_{i=1}^n \langle \sigma_i, \tau_i \rangle,$$

$$\text{where } \sigma_1 \otimes \cdots \otimes \sigma_n, \tau_1 \otimes \cdots \otimes \tau_n \in \bigotimes_{i=1}^n C_*(K_i) \quad \langle \sigma, \tau \rangle = \begin{cases} 1, & \sigma = \tau; \\ 0, & \text{otherwise.} \end{cases}$$

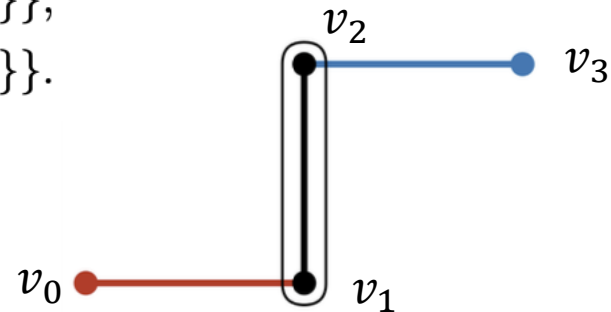
The p th interaction Laplacian $\Delta_p : IC_p(\{K_i\}_{1 \leq i \leq n}) \rightarrow IC_p(\{K_i\}_{1 \leq i \leq n})$

$$\Delta_p := (d_p)^* \circ d_p + d_{p+1} \circ (d_{p+1})^*, \quad p \geq 0.$$

$$L_p = B_p B_p^T + B_{p+1}^T B_{p+1}.$$

Proposition 2.3. $IC_p(\{K_i\}_{1 \leq i \leq n}) = \ker \Delta_p \oplus \text{im } d_{p+1} \oplus \text{im } (d_p)^*$. Here, $\ker \Delta_p$ is isomorphic to the interaction homology $H_p(\{K_i\}_{1 \leq i \leq n})$.

Example 2 $K_1 = \{\{v_0\}, \{v_1\}, \{v_2\}, \{v_0, v_1\}, \{v_1, v_2\}\},$
 $K_2 = \{\{v_1\}, \{v_2\}, \{v_3\}, \{v_1, v_2\}, \{v_2, v_3\}\}.$



	$\{v_0, v_1\} \otimes v_1$	$\{v_0, v_1\} \otimes \{v_1, v_2\}$	
$v_1 \otimes v_1$	$\{v_1, v_2\} \otimes v_1$		
	$\{v_1, v_2\} \otimes v_2$	$\{v_1, v_2\} \otimes \{v_1, v_2\}$	
$v_2 \otimes v_2$	$v_1 \otimes \{v_1, v_2\}$		
	$v_2 \otimes \{v_1, v_2\}$	$\{v_1, v_2\} \otimes \{v_2, v_3\}$	
	$v_2 \otimes \{v_2, v_3\}$		
$\rightarrow IS_*(\{K_1, K_2\})$			
$IS_0(K_1, K_2)$	$IS_1(K_1, K_2)$	$IS_2(K_1, K_2)$	

$$L_0 = B_1 B_1^T = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \quad L_2 = B_2 B_2^T = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

$$L_1 = B_1^T B_1 + B_2 B_2^T = \begin{pmatrix} 3 & -1 & 0 & 0 & 0 & 1 \\ -1 & 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 1 & 0 & 0 & 0 & -1 & 3 \end{pmatrix},$$

$$\text{Spec}(L_0) = \{3, 3\};$$

$$\text{Spec}(L_1) = \{0, 3 - \sqrt{3}, 2, 3, 3, 3 + \sqrt{3}\};$$

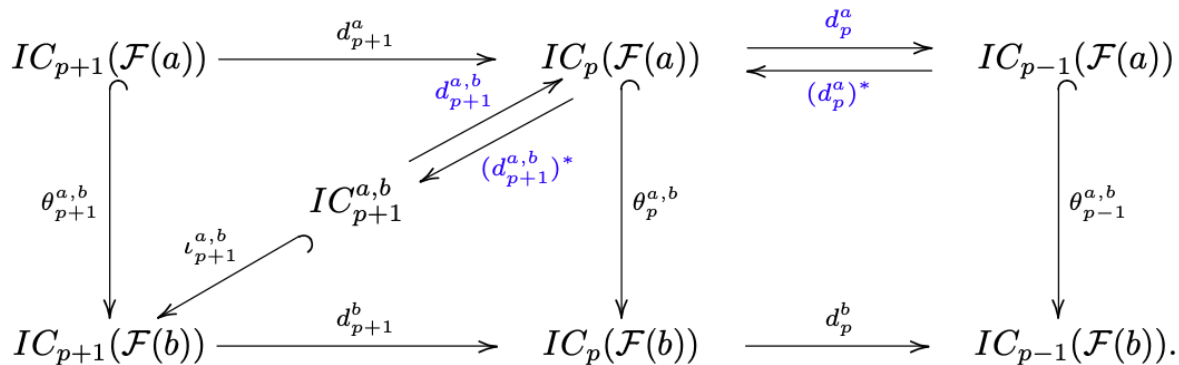
$$\text{Spec}(L_2) = \{3 - \sqrt{3}, 2, 3 + \sqrt{3}\}.$$

$$\beta_0^2(\{K_1, K_2\}) = 0$$

$$\beta_1^2(\{K_1, K_2\}) = 1$$

$$\beta_2^2(\{K_1, K_2\}) = 0$$

Persistent Interaction Laplacian



The p -th (a,b) -persistent interaction Laplacian $\Delta_p^{a,b} : IC_p(\mathcal{F}(a)) \rightarrow IC_p(\mathcal{F}(a))$ is defined by

$$\Delta_p^{a,b} := d_{p+1}^{a,b} \circ (d_{p+1}^{a,b})^* + (d_p^a)^* \circ d_p^a.$$

Where will interaction homology be a good tool?

1. Multi-component systems

- protein–ligand complexes, host–guest chemistry, multi-cluster nanostructures (like carboranes, metallofullerenes)

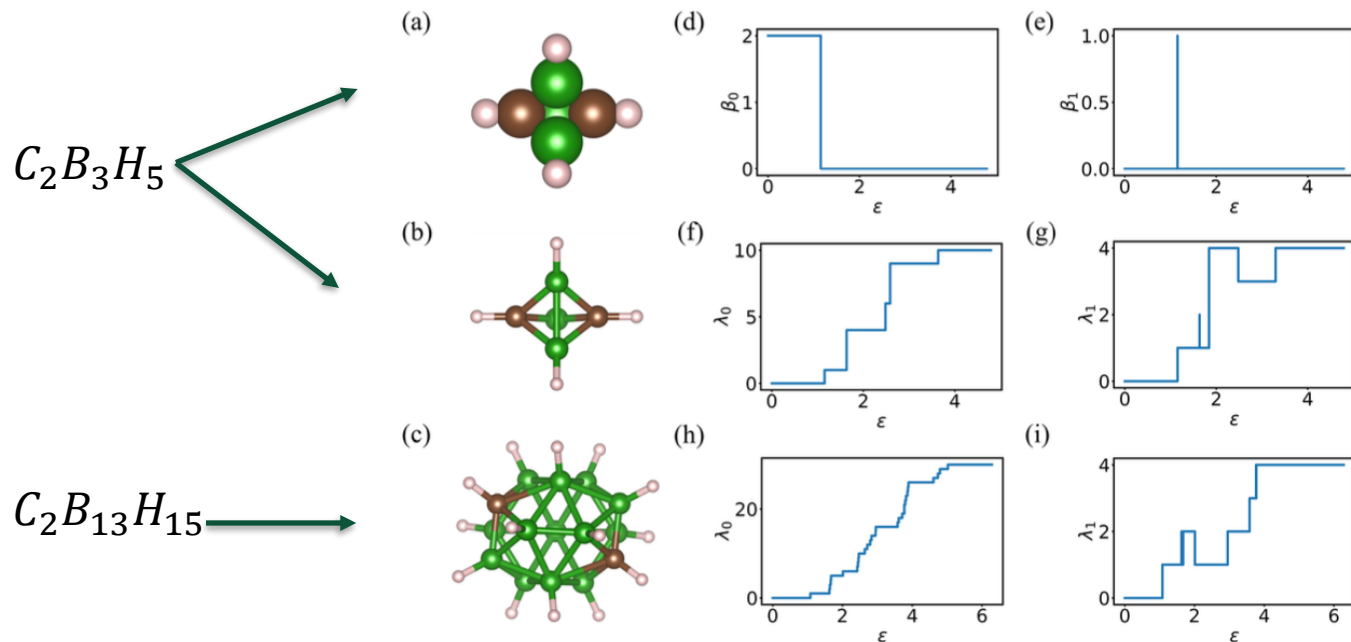
2. Social or communication networks

- bipartite or multi-layer graphs (users $\leftrightarrow$ products, authors $\leftrightarrow$ topics)

3. Multi-brain or multi-sensor neural data

- hyper scanning EEG (brain–brain interaction), multi-sensor fusion

Structural Analysis of Closo-Carboranes

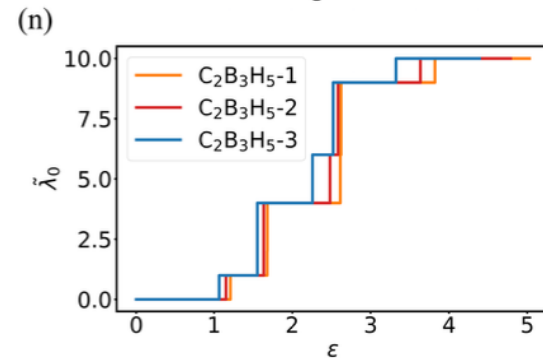
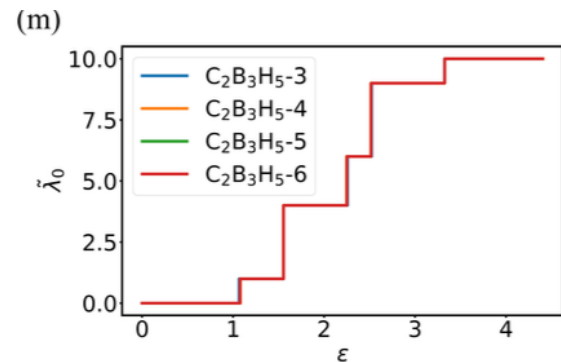
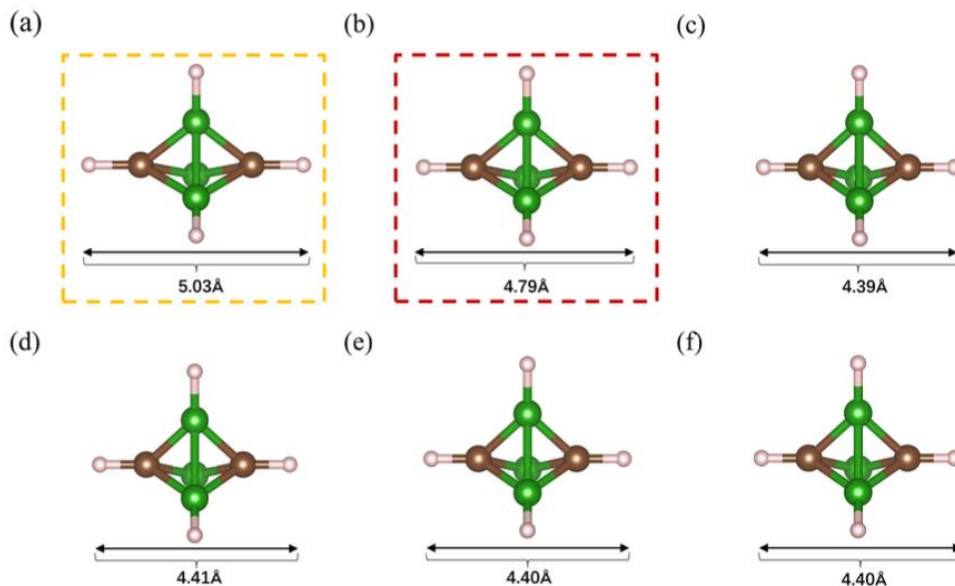


Interaction between
 $\{C_2B_3\}, \{C_2H_5\}$

$\{C_2B_{13}\}, \{C_2H_{15}\}$

Different Configurations

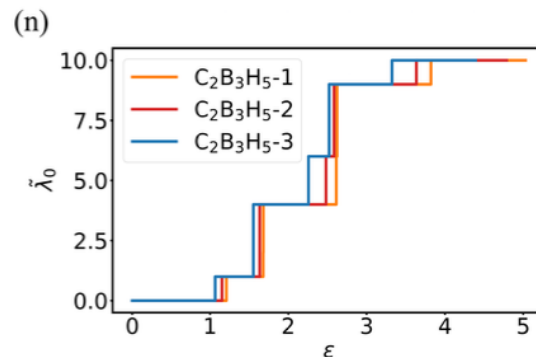
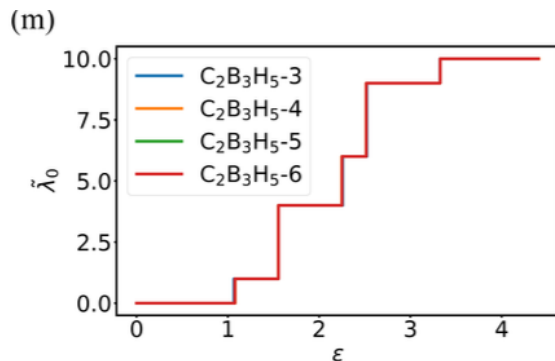
$C_2B_3H_5$ has 6 possible configuration where $C_2B_{13}H_{15}$ has 135 configurations



6 different configuration of $C_2B_3H_5$

Relative enthalpy $\Delta E = E(X) - E(C_2B_3H_5-5) = E(X) - 153.810980788$.

$C_2B_3H_5$	$C_2B_3H_5-1$	$C_2B_3H_5-2$	$C_2B_3H_5-3$	$C_2B_3H_5-4$	$C_2B_3H_5-5$	$C_2B_3H_5-6$
ΔE	0.078485	0.033149	0.000178	0.000026	0.0	0.0



Comparison of chlorophyll-a and chlorophyll-b

1. **Chlorophyll** is a green pigment found in plants, algae, and some bacteria. It enables **photosynthesis**, converting light energy into chemical energy.
2. In our model, we define two **interacting components**:
 - {C, H, O, N} — structural and energy-propagation part
 - {N, Mg} — energy-absorption and generation center
3. **Chlorophyll-a** ($C_{55}H_{72}O_5N_4Mg$) and **Chlorophyll-b** ($C_{55}H_{70}O_6N_4Mg$) differ mainly by an extra carbonyl group in chlorophyll-b.

